Chulalongkorn University · Department of Chemical Engineering

Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator

Electrical load list, cable sizing and hazardous-area schedule

NDFH-V1-EE-LL-001 · Rev 0 · Issued for teaching

Rev	Date	Description	Originator	Checker	Approver
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Title block	
Client	Chulalongkorn University · Department of Chemical Engineering
Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator
Document no.	NDFH-V1-EE-LL-001
Title	Electrical load list, cable sizing and hazardous-area schedule
Drawings	PFD-100 rev 1; P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev E (EPC alternative)
Scope	Electrical consumers of H-100, maximum demand, motor cable sizing, UPS for the SIS and DCS, hazardous-area schedule.

All numbers in this document come from the Natural Draft Fired Heater Ver 1 simulator model (design case and model runs) or are stated as assumed. It is a teaching document, not a design for construction.

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1 · Load list

Tag	Consumer	Type	Rated kW	Absorbed kW	LF	Category	kW in	kVA	FLC A
P-100A	Feed pump A (running)	Motor	15	11.11	0.74	continuous	12.07	14.04	27.4
P-100B	Feed pump B (spare)	Motor	15	11.11	0.74	standby	12.07	14.04	27.4
IGN-101	Igniter transformer	Transformer	0.3	0.30	1.00	intermittent	0.30	0.33	—
HT-201	Fuel oil line heat tracing (A7)	Heater	2	2.00	1.00	continuous	2.00	2.00	—
UPS-SIS	SIS / BMS logic solver and scanner (UPS)	UPS	1.5	1.20	0.80	continuous	1.20	1.33	—
UPS-DCS	DCS share, transmitters (UPS)	UPS	1	0.80	0.80	continuous	0.80	0.89	—
AT-026/028/029	Analyzers (heaters, sample)	Analyzer	1.1	1.10	1.00	continuous	1.10	1.22	—
LT-100	Lighting and small power, heater area	Lighting	3	2.40	0.80	continuous	2.40	2.67	—

2 · Maximum demand

Category	Σ kVA	Factor	Demand kVA
Continuous	22.2	1	22.2
Intermittent	0.3	0.3	0.1
Standby	14.0	0.1	1.4
Maximum demand			**23.7**

3 · Cable sizing, P-100A/B motors

15 kW, 400 V, FLC 27.4 A, DOL start 6 × FLC at pf 0.3, route 150 m, derating 0.8 (grouped on tray). Copper, resistance at 70 °C, ratings typical of IEC 60364-5-52 tables (A10). Selected: **10 mm²**.

Size	Rating × 0.8 ≥ FLC	Running ΔV ≤ 5 %	Starting ΔV ≤ 15 %
2.5 mm²	22 A ✗	13.66 % ✗	29.3 % ✗
4 mm²	29 A ✓	8.53 % ✗	18.5 % ✗
6 mm²	37 A ✓	5.73 % ✗	12.7 % ✓
10 mm²	50 A ✓	3.44 % ✓	7.9 % ✓
16 mm²	68 A ✓	2.18 % ✓	5.2 % ✓
25 mm²	90 A ✓	1.40 % ✓	3.6 % ✓
35 mm²	110 A ✓	1.04 % ✓	2.8 % ✓
50 mm²	134 A ✓	0.78 % ✓	2.3 % ✓

4 · Power for the SIS and DCS

The SIS is de-energise-to-trip: XV-301 and XV-302 close when their solenoids lose power. The logic solver, solenoids and flame scanner (1.5 kVA) and the DCS share (1.0 kVA) run from a UPS; a 60 Ah, 110 V battery gives about 141 min at 80 % usable capacity, against 30 min for an orderly shutdown.

5 · Hazardous-area schedule

After IEC 60079-10-1 and API RP 505; fluid properties assumed (A7, A8). The heater is itself a source of ignition: classification does not protect against a leak of hot feed above its auto-ignition temperature.

Source	Fluid and condition	Grade	Why	Zone and extent	Group	T-class
P-100A/B seals	Feed, 40 °C (A8: flash point 45 °C)	Secondary	Below flash point at the pump: no flammable vapour expected; Zone 2 kept for mist on a seal failure	Zone 2, 3 m radius	IIA	T3
E-100 flanges, tube side outlet	Feed at 180 °C, above flash point	Secondary	Leak gives vapour	Zone 2, 3 m	IIA	T3
FV-010, coil inlet flanges	Feed at 180 °C	Secondary	Leak gives vapour	Zone 2, 3 m	IIA	T3
Heater outlet 3"-P-105 flanges	Feed at 350 °C, above AIT 250 °C (A8)	Secondary	A leak can ignite on its own: classification does not prevent it; fire protection and EIVs do	Zone 2, 3 m (and fire zone)	IIA	T3
Fuel oil train (FV-300, XV-301/302/303)	Fuel oil at 120 °C, above flash point 66 °C (A7)	Secondary	Leak and vent from XV-303 to a safe place	Zone 2, 3 m; vent outlet Zone 1, 1.5 m	IIA	T3
Burner front and firebox	Open flame	—	Source of ignition, not of release; keep electrical equipment out of the burner-front zone or rate it	Unclassified by convention (ignition source)	—	—

Assumptions used in this document

No.	Assumed	Value	Used in
A1	Feed viscosity	1.0 mPa s at 40 °C, 0.5 mPa s at 180–350 °C (Design tab value)	Line sizing
A2	Feed vapour pressure at 40 °C	5 kPa(a)	NPSH available
A3	Pump efficiency P-100A/B	60 % at rated flow	Absorbed power, motor size
A4	Heating fluid from Unit A	hot oil, ρ 850 kg/m³, c_p 2.3 kJ/kg K, return 150 °C, operating 600 kPa(g), design 1000 kPa(g) / 260 °C	E-100 rating, tube-rupture case
A5	E-100 overall U	350 W/m² K (liquid–liquid, clean), 15 % area margin	E-100 area
A6	E-100 geometry	TEMA AES, ¾ in 14 BWG tubes (ID 14.8 mm), shell OD 0.61 m × 6.1 m	Wetted area, rupture area
A7	Fuel oil	ρ 900 kg/m³, μ 10 mPa s, delivered heated to 120 °C, flash point 66 °C	Fuel line, area classification
A8	Feed fluid	flash point 45 °C, auto-ignition 250 °C, latent heat 300 kJ/kg, vapour M 150, k 1.05 at relief	Relief and area classification
A9	Allowable stresses	illustrative ASME II-D values (see the stress table); confirm from the code edition in force	Wall thickness
A10	Electrical	400 V, 3-phase, 50 Hz; motor η 92 %, pf 0.86; starting current 6 × FLC	Load list, cable sizing
A11	Lead times	typical market lead times in weeks, for teaching	Procurement schedule
A12	Corrosion allowance	3 mm carbon steel coil and piping, 3 mm vessels	Wall thickness

6 · Checks

Check	This document	Compared with	Result
P-100 absorbed power at design from the model	11.11 kW	11.11 kW (live PI-002, A3)	✓
Motor ≥ 1.1 × highest absorbed power on the curve	15 kW	12.4 kW	✓
SIS and BMS loads on UPS	UPS-SIS 1.5 kVA	de-energise-to-trip XV-301/302	✓
T-class surface limit below the fluid auto-ignition temperature	T3: 200 °C	AIT 250 °C (A8)	✓
Heater outlet above auto-ignition (leak ignites on its own)	350 °C	250 °C (A8)	info